COURSE TITLE : ENVIRONMENTAL ENGINERING
COURSE CODE : 4071
COURSE CATEGORY : A
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 56
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Water Pollution & Control	14
2	Air Pollution & Control	14
3	Noise Pollution & its Control	14
4	Solid Waste Management	14
TOTAL		**56**

COURSE OUTCOME :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	Understand the effect of water pollution
	2	Comprehend source of water pollution
	3	Appreciate various methods of waste water treatment
2	4	Understand effect of air pollution and its control
	5	Understand effect of noise pollution & its control
	6	Comprehend the various solid waste control methods
3	7	Understand the noise measurement and control
	8	Understand the pollution due to the Radio Activity and its control
	9	Understand the source of soil pollution and control Understand the effect of green chemistry Understand the effect of E-waste

SPECIFIC OUTCOME

MODULE I WATER POLLUTION AND CONTROL

1.1.0 Understand the effects of water pollution

1.1.1 Define water pollution
1.1.2 Explain the sources of water pollution
1.1.3 Describe the effects of water pollution on human being
1.1.4 Describe the effects of water pollution on animals

1.2.0 Understand the characterization of waste water

1.2.1 List the characteristics' of waste water
1.2.2 Define B.O.D
1.2.3 Define C.O.D
1.2.4 Explain the procedure for determination of BOD
1.2.5 Explain the procedure for determination of COD

1.3.0 Understand the various methods of waste water treatment

1.3.1 List the various steps of preliminary treatment
1.3.2 Illustrate the preliminary treatment of waste water
1.3.3 Select the important operation in primary treatment
1.3.4 Identify the Aids to sedimentation process
1.3.5 State mechanical flocculation
1.3.6 Define coagulation
1.3.7 List the different types of coagulants
1.3.8 Illustrate the secondary treatment processes of waste water
1.3.9 State the aerobic effluent treatment
1.3.10 List the types of aerobic process
1.3.11 Illustrate the working of Tricking filter
1.3.12 Draw the figure and name the parts of Tricking Filter
1.3.13 Explain the activated sludge process
1.3.14 Identify the Oxidation ditch
1.3.15 Illustrate the function of Oxidation pond
1.3.16 State the effects of nutrients in the oxidation pond
1.3.17 Explain the tertiary treatment processes of waste water
1.3.18 list the operations/equipment used in the tertiary treatment
1.3.19 Illustrate the activated carbon adsorption method
1.3.20 Explain sewage/sludge treatment and disposal
1.3.21 List the various steps in the sludge treatment
1.3.22 Illustrate the sludge Digestion by anaerobic process
1.3.23 list the advantages of anaerobic process

MODULE II AIR POLLUTION AND CONTROL

2.1.0 Understand the effects of Air pollution

2.1.1 Define air pollution
2.1.2 Explain the effects of air pollution on materials
2.1.3 Describe the effects of air pollution to human beings
2.1.4 Describe the effects air pollution to animals

2.2.0 Understand the sources and classification of air pollutants

2.2.1 List the major air pollutants from industries
2.2.2 Identify the sources of gaseous air pollutant
2.2.3 State the sources of dust, fumes, smoke and mist

2.3.0 Understand the Air pollution sampling and measurement

2.3.1 State ambient and stack gas sampling
2.3.2 List the methods for monitoring air pollutant
2.3.3 Illustrate the typical air sampling methods
2.3.4 List the methods for collection of gaseous air pollutants from ambient atmosphere

2.3.5 Illustrate the absorption process of gaseous pollutant into a liquid medium
2.3.6 Describe the procedure for collection of particulate pollutants
2.3.7 Illustrate dust fall jar collection for particulate measurements
2.3.8 Illustrate High volume sampler for particulate measurements
2.3.9 Illustrate the collection of gaseous pollutant from stack
2.3.10 Explain how you can find out the amount of particulate matter of gaseous sample from stack

2.4.0 Understand the air pollution control methods and Equipments
2.4.1 Explain different air pollution control methods
2.4.2 List the basic mechanism of removing particulate matter from gas streams
2.4.3 Explain the working of following particulate control equipments
a) Gravitational settling chambers
b) Cyclone separators
c) fabric filters/bag filters
d) Electrostatic precipitators
e) Wet collectors(venture scrubbers)

MODULE III NOISE POLLUTION & CONTROL

3.1.0 Understand the sources and the effects of Noise Pollution
3.1.1 Define "Noise" pollution
3.1.2 List the various "Fifteen sound sources" and their decibel scale
3.1.3 List the physical and psychological effects of noise pollution
3.1.4 Explain the psychological effect of noise pollution

3.2.0 Understand the noise measurement and control
3.2.1 Deduce the noise control program in industries
3.2.2 List the equipments used for noise measurement
3.2.3 List the type of noise from equipments in industries
3.2.4 List the type of control program in industries
3.2.5 State administrative control of noise
3.2.6 List the engineering methods for control of noise
3.2.7 List the sources of noise in the industries
3.2.8 List the various method of path control of noise
3.2.9 Illustrate the function of " Barriers"
3.2.10 Illustrate the function of mufflers or silencers
3.2.11 List the personal protective equipment used in indutries
3.2.12 State TLV, SIL, NEI

3.3.0 Understand the pollution due to the Radio Activity and its control
3.3.1 Describe the source of radioactive pollution
3.3.2 Explain the effects of radioactive pollution
3.3.3 Explain the monitoring of radioactive pollution
3.3.4 Describe the control of radioactive pollution
3.3.5 Explain radiation effect to the human body from radioactive wastes

MODULE IV SOLID WASTE MANAGEMENT

4.1.0 Understand the source of soil pollution and control

4.1.1 List the basic composition of earth
4.1.2 Explain the different steps of earth crust
4.1.3 List the three composition of the soil
4.1.4 Explain the sources of soil pollution
4.1.5 Describe effects of soil pollutants on Human beings and plants
4.1.6 Explain the effects of industrial effluent on land
4.1.7 Describe the control method of soil pollution

4.2.0 Understand the effect of green chemistry

4.2.1 Define green chemistry
4.2.2 List the Ten principles of Green chemistry
4.2.3 List the objectives of environmental management
4.2.4 List the environmental protection act

4.3.0 Understand the Solid waste management and control

4.3.1 List the sources of solid waste
4.3.2 List the methods of solid waste treatment and disposal
4.3.3 Explain the following waste treatment method
1) Composting
2) Sanitary land filling
3) Thermal process including incineration and pyrolysis
4) Recycling and reuse

4.4.0 Understand the effect of E-waste

4.4.1 List the sources of e-waste
4.4.2 List the constituents of e-waste
4.4.3 Identify the health effect of e-waste
4.4.4 Deduce the method of e-waste management

CONTENT DETAILS

MODULE I WATER POLLUTION & CONTROL

Sources of water pollution – effects of water pollution – control standards of KSPCB – BOD, COD determination – primary, secondary, Tertiary waste water treatment – aerobic and anaerobic digestion – activated sludge process – activated carbon process – sewage treatment

MODULE II AIR POLLUTION AND CONTROL

Definition of air pollution – sources – effects of air pollution on man, material and animals- sources of gaseous air pollutant- sources of dust, fumes, smoke and mist
-control of dust emissions – Electrostatic Precipitators, Bag filters, cyclone separators, Venturi scrubbers, etc – sources of gaseous pollutants – absorption process of gaseous pollutant into a liquid medium-stack gas sampling- typical air sampling methods- collection of gaseous air pollutants from

ambient atmosphere- dust fall jar collection for particulate measurements- High volume sampler for particulate measurements- collection of gaseous pollutant from stack- basic mechanism of removing particulate matter from gas streams- working of pollutants- Gravitational settling chambers- Cyclone separators-
fabric filters/bag filters-Electrostatic precipitators- Wet collectors(venture scrubbers)

MODULE III NOISE POLLUTION AND CONTROL

Radioactive pollution and Control

Noise pollution – source of sound and their decibel scale – physical and psychological effects of noise pollution –Noise control criteria- noise control program in industries
– equipments used for noise measurement-noise pollution control program in industries – administrative& engineering methods for control of noise-sources of noise in the industry- various method of path control of noise- function of " Barriers", mufflers or silencers- personal protective equipment used in industries- State TLV, SIL, NEI

Source of radioactive pollution- effects of radioactive pollution- monitoring of radioactive pollution-control of radioactive pollution- effect to the human body from radioactive wastes

MODULE IV SOLID WASTE MANAGEMENT & CONTROL

List the basic composition of earth- different steps of earth crust- composition of the soil- sources of soil pollution- effects of soil pollutants on Human beings and plants- effects of industrial effluent on land- the control method of soil pollution- green chemistry- Ten principles of Green chemistry- objectives of environmental management-- E.I.A and EIS- environmental protection act- the sources of solid waste-methods of solid waste treatment and disposal- Composting- Sanitary land filling-Thermal process including incineration and pyrolysis- Recycling and reuse

REFERENCES

S.S.Dara	-	Text book of environmental chemistry and pollution control
N.Manivasakam	-	Environmental Pollution
C.S.Rao	-	Environmental Engg. and Pollution
IIT CHENNAI	-	Chem Tech Vol-1